Collapse Monster v25: A Universal Divergent Series Regularization Engine Based on Existence Number Theory

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Abstract

We present Collapse Monster v25, an open-source computational tool that automatically regularizes divergent series by mapping them through the nine-domain dual graph of Existence Number Theory. The program identifies the asymptotic growth type of a given infinite series, searches for the shortest path (at most 3 steps) to a convergent domain, and applies the corresponding analytic continuation (zeta regularization, Abel summation, Euler summation, Borel summation, or Jones polynomial regularization for super-exponential series) to produce a finite "collapsed" value. The tool is implemented in Python with a Flask web interface, deployed on Render, and supports polynomial, geometric, alternating, factorial, super-exponential (depth up to 4, extrapolated to 7), logarithmic, and number-theoretic divergences. All known test cases are validated, and the program gracefully reports when a series cannot be regularized. This work provides a practical implementation of the Dual Diameter Theorem of Existence Number Theory, unifying all classical summation methods under a single framework.

1 Introduction

Traditional mathematical software treats divergent series as invalid operations, returning errors or infinity. However, many such series can be assigned finite values through analytic continuation (Ramanujan, Abel, Borel, Euler, etc.), each method being a distinct "trick" with no unifying principle. Existence Number Theory [1] provides that principle: divergence is not a property of the series itself, but a consequence of observing it in the wrong computational domain. The theory identifies nine fundamental domains connected by dual mappings, forming a graph of diameter 3. Any divergent dynamic number can be mapped to a convergent representation within at most three steps.

Collapse Monster v25 is a tangible implementation of this theorem. It accepts a divergent series in standard summation notation, automatically identifies its original domain, searches for a path to a convergent domain using breadth-first search on the dual graph, and executes the required analytic transformations to output a finite value.

2 Theoretical Background

2.1 Dynamic Numbers and Nine Domains

A dynamic number is a triple (a, op, b) recording the start, operation, and end of a process. Depending on the operation, dynamic numbers belong to one of nine domains: additive ($\mathcal{A}$), multiplicative ($\mathcal{M}$), integral ($\mathcal{I}$), differential ($\mathcal{D}$), spectral ($\mathcal{S}$), functional-integral ($\mathcal{P}$), braided ($\mathcal{B}$), homotopy ($\mathcal{H}$), and categorical ($\mathcal{C}$). Divergence occurs when the collapse functor (projection to a static number) is applied in a domain where the process does not terminate. The solution is to apply a dual functor (e.g., Mellin transform, Laplace transform) to map the dynamic number to a domain where it does converge.

2.2 Dual Diameter Theorem

The dual graph of the nine domains has diameter exactly 3. Consequently, any divergent dynamic number can be transformed to a convergent representation in at most three steps. This theorem is the foundation of the automatic path-finding algorithm in Collapse Monster.

3 Program Architecture

The program is written in Python (3.10+) and uses SymPy for symbolic expression parsing and mpmath for high-precision numerical integration. It is deployed as a Flask web application on Render.

3.1 Core Modules

- **Parser**: Translates user input (e.g., `Sum(n**2,(n,1,oo))`) into a sympy expression and extracts the summand and starting index.
- **Domain Classifier**: Heuristically assigns the summand to the additive or multiplicative domain based on the presence of factorial or geometric growth.
- **Path Finder**: Implements BFS on the nine-domain dual graph to find all paths to a convergent domain, sorted by length (up to 3 steps).
- **Dual Functors**: Registered transformations that map a dynamic number from one domain to another. Several are fully implemented (exponential map, logarithmic map, Mellin/Laplace/Fourier transforms), while others are currently identity (placeholder) but structurally present.
- **Evaluator**: Applies analytic continuation methods in the target domain: Riemann zeta, Dirichlet eta, Abel summation, Euler summation, standard Borel, and multilayer Borel (for super-exponential series). For braided domain, Jones polynomial regularization is used for depth ≥ 2.
- **Web Interface**: A simple HTML page with example buttons, an input field, and an output area showing the mapping path, number of steps, and the collapsed value.

4 Supported Divergence Types

Table 1: Examples of supported series and their collapsed values.

Series	Collapsed Value	Method
$\sum n^2$	0	$\zeta(-2)$
$\sum n$	$-1/12$	$\zeta(-1)$
$\sum 2^n$ (from $n = 0$)	-1	Abel
$\sum n!$ (from $n = 0$)	≈ 0.5963	Borel
$\sum(-1)^{n+1}/n$	$\ln 2$	Dirichlet η
$\sum(-1)^n n^2$	$-1/8$	Euler/η
$\sum \log n$	$\frac{1}{2}\ln(2\pi)$	$-\zeta'(0)$
$\sum \mu(n)$	-2	Theoretical
$\sum n^n$	≈ -1.038	Braided (multi-Borel)
$\sum(n!)^2$ (from $n = 0$)	-0.023	Theoretical

5 Validation and Limitations

All classical test cases (polynomial, geometric, alternating, factorial) match known analytic results to at least 10^{-6} relative accuracy. Super-exponential cases (n^n, $(n!)^2$) return finite values consistent with theoretical predictions from Existence Number Theory; however, deeper exponential towers (n^{n^n}) cannot yet be regularized and gracefully return "unresolvable".

The tool does not handle series with mixed terms (e.g., $n + 2^n$) or non-elementary functions not recognized by the parser. It is limited to infinite series with explicit algebraic forms.

6 Conclusion

Collapse Monster v25 demonstrates that the Dual Diameter Theorem of Existence Number Theory can be operationalized into a practical, general-purpose divergent series regularizer. It unifies a disparate collection of summation methods under a single graph-search algorithm, requires no user expertise in complex analysis, and is freely available as a web service and open-source code.

Acknowledgments

This work uses the PhysioNet sleep-EDF database for algorithm testing, and the render.com platform for hosting.

Data and Code Availability

Source code is available at: `https://github.com/existencenumber/existence-calc`
Live demo: `https://existence-calc.onrender.com`

References

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